

Do AI Platform Agents Know What They Must Not Forget?

Compaction-Aware Intervention Before State-Changing Operations

ARIN 5203 Course Project Proposal

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Long-running tool-using agents often shorten histories to control context length, latency, and cost, but compaction can omit an approval boundary, prohibition, expiry, entity, amount, or recovery condition that determines whether a later tool call is safe. We will test whether a post-compaction evidence audit, applied immediately before a state-changing operation, can use missing or conflicting evidence to choose Execute, Confirm, or Handoff under a fixed context budget. The project builds on stateful agent evaluation in τ -bench and ToolSandbox, mutation-gated verification in SABER, evidence-preserving context management in U-Fold, and the constraint-loss problem demonstrated by Governance Decay. Our intended contribution is not another general-purpose compressor; it is a selective intervention layer that links compaction failures to action-time control. We will construct 20 base scenarios in a deterministic AI-platform simulator, covering quota changes, workload preemption, resource transfer, model rollout, and capacity reclamation. Each base scenario will produce three matched variants—complete evidence, one action-critical item removed, and matched non-critical noise removed—for 60 cases, with sibling variants kept in the same split. Evidence will cover the target entity, requested scale, authorization scope, validity or expiry, negative constraints, and reversibility or recovery. We will compare full history, a recent-window baseline, generic summarization, fixed constraint pinning, a U-Fold-inspired intent-summary-plus-tool-log baseline, and our audit-guided policy. The audit will check whether required evidence is present and mutually consistent; sufficient evidence permits Execute, an answerable omission triggers Confirm, and a policy conflict or unresolved high-impact action triggers Handoff. A state-transition oracle will score consequences independently of the router. Thresholds will be selected on development cases and frozen before testing. We will report harmful-mutation rate, safe autonomous completion, critical-evidence retention, intervention accuracy, human-review burden, context reduction, latency, and token cost, together with ablations and error analysis. Qingcheng Yu will lead scenario and intervention specifications, drawing on agent lifecycle and human-in-the-loop evaluation experience. Siu Kwun Yeung will lead evidence and audit design, drawing on enterprise multi-agent verification and human review. Qingyin Guo will lead simulator and tool integration, drawing on multi-source agent planning and API orchestration. A small local pilot will validate feasibility only; we will not present synthetic results as production guarantees.